Engineering Design Specification



Team 2

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1.0 Executive Summary

This document overviews the engineering design specifications for Team 2 in UCF's CRASH competition. The first section will introduce the competition and overview the goals of team 2. The second section will detail the specifications set by the stakeholder, UCF, for the competition, and then break down how requirements were derived from these specifications. Finally, a comprehensive requirements table detailing all requirements, including those set by competition rules, as well as requirements derived from those initial needs that were established by the team.

1.1 Problem Statement

The CRASH competition fundamentally asks: How can you deliver a fragile payload safely? Maintaining safety through all phases of flight, ascent, descent, and recovery, without structural failure or material damage? The system design problem is to develop an L1-class rocket capable of transporting a payload of eggs and water while maintaining their structural integrity and containment throughout the flight. The payload must endure intense flight conditions, vibrational, thermal, and impact loads related to launch, landing, and conditions that mirror the stress faced by sensitive payloads in the professional aerospace industry.

The design problem extends beyond the typical rocketry competition goal of reaching a high altitude. It requires integration of propulsion, recovery, and payload protection subsystems into a cohesive architecture that prioritizes payload survivability. The payload bay must incorporate shock absorption, vibration isolation, and mechanical restraint systems, while the recovery team must provide a controlled descent that minimizes the rocket's ground impact. Achieving a balance between maximizing points earned in the competition and with survivability of the payload.

The primary stakeholder in the CRASH competition which will evaluate the team's ability to deliver the eggs and water safely under the competition's constraints. The CRASH competition aims to challenge students to find the balance between point maximization, safety, and meeting minimum altitude. Parallel to industry finding a balance between their own needs and constraints, typically cost and performance. The CRASH competition also puts a strong emphasis on safety, the top priority being the safety of the payload and water, demonstrated by the automatic disqualification of rockets impacting above a speed limit of 25 m/s. The broader relevance of the rocket is the delivery of fragile payloads, common in the real-world space industry. Applications like the transport of biological materials, microgravity science payloads, and precision instrumentation, where the survival of the fragile payloads determines the success of the mission. The CRASH Rocket will contribute to the improvement of payload packaging standards and safe recovery methods.

The team's current efforts are focused on defining user needs, translating them into measurable engineering requirements, and conducting system-level trade studies to evaluate the current designs in the industry. The ultimate objective to deliver an L1-class rocket system that maximizes payload survivability, a mission-driven rocket and payload design to meet the target.

1.2 Objectives Overview

Team 2 is committed to competing for first place in the CRASH competition. In order to have the best odds of achieving this goal, it will be important to ensure the base requirements of the stakeholder are well within the set specifications, such that the primary design efforts can be focused upon maximizing points scored. The team also would like to design a vehicle that is aesthetically and functionally pleasing for the competition. This goal is set so that the team can be sure that their design is something they can be proud of when showing to peers.

2.0 Engineering Requirements and Preliminary Specifications

For the UCF CRASH competition, there is a set of defined requirements that the project must meet in order to be valid for scoring. In addition to the necessary requirements, there are optional objectives that can be achieved to score more points. This section will break down the specifications necessary for validation into needs, and the optional specifications into wants.

2.1. Specification Priority

As there are no direct third party users of the rocket system, all priorities are derived by the competition rules. All “threshold” goals of the competition are considered “needs”, as if they are not met, the team is disqualified. All goals that offer bonus points on top of threshold goals will be designated “wants”.

1. [Need] Safe Payload delivery to ground (little to no damage: no visible cracks or dents on the eggs, no visible cracks or leaks for the water container). The contest rules require the payload is recovered with at least two eggs fully intact, as well as no leakage having occurred in the water vessel.
2. [Need] Landing within a reasonable distance of the launch zone. The contest rules require the vehicle to be recovered within 800 feet from the launch pad.
3. [Need] Vehicle is recovered in flyable condition. The contest rules require the vehicle to be recovered in a flyable condition.
4. [Need] Undergo a controlled flight (Maximum altitude and landing speed). The contest rules require the descent velocity during the final five seconds of flight to not exceed 25 feet per second, as well as an apogee requirement of between 999 feet and 1500 feet.
5. [Need] Diversity of cargo (eggs and water). The contest rules require a minimum of 2 eggs and 5 oz of water on board.
6. [Need] Ease of access to cargo (time requirements). The contest rules require a payload loading and unloading time of 5 minutes or less.

7. [Need] Timed ejection of recovery equipment. Ejection of rockover equipment in a dual deployment manner gives the team more points and gives the team control over the speed of descent and the distance of landing.

8. [Need] Logging of flight data such as velocity, acceleration and altitude. Logging of flight data is needed to prove apogee requirements are met on launch day. A tracking system also gives the team additional points.

9. [Need] Backup equipment to ensure data recoverability. Allows the opportunity to be scored even if the primary electronics fail.

10. [Need] Use of sufficiently scaled motor for the task ($\leq$ 320 Ns). The contest rules require the sum of all launch vehicle propulsion systems to contain no more than 320 Newton seconds of total impulse.

11. [Want] Deployment of a parachute during recovery.

12. [Want] Capability of carrying more cargo. To maximize points scored in the competition, the vehicle should be capable of carrying additional eggs and water.

13. [Want] Some form of acoustic or GPS locating system. Additional points will be earned from carrying a location/tracking system.

14. [Want] Employment of a dual-deployment recovery system. Additional points will be earned from the utilization of a dual-deployment recovery system.

15. [Want] Inclusion of a functional camera on the rocket.

16. [Want] Video playback of camera footage.

17. [Want] Live stream video footage from rocket to ground station. Additional points will be earned from the transmission of live video footage from the vehicle to a ground station.

2.2. Benchmarking Requirements

Payload packaging:

Normal (not in a rocket) egg packaging is typically done in 3 ways. Firstly, packaging eggs in rice husks, wheat chaff ,or chopped straw, in a crate. Secondly, packaging in filler trays made of wood pulp or plastic , constructed to stack on each other. Thirdly, packaging in smaller casings made from polystyrene. It is noted that polystyrene is superior in cushioning. [1] The standard of performance in the industry is minimizing the quantity of cracked eggs due to jolts and vibration.

In the aerospace industry, protection for payloads typically falls under two methods, cushioning with foam, gel, and honeycombs, or suspension with elastic cords and spring mounts. [2] The standard in industry is the total protection of payloads from the conditions caused by the rocket's ascent and descent, or ascent and delivery to space.

Benchmarks from the egg transport and aerospace payload industry give us these requirements:

1. The protection system shall reduce vibration
2. The protection system shall reduce impact forces

Acknowledging the established industry benchmark, we now define the standard derived from our project:

1. The egg protection system shall reduce vibration intensity to less than or equal to 7.5 m/s^2.
2. The egg protection system shall limit impact forces to less than 35N.
3. The egg protection system shall allow for eggs to be loaded and unloaded within 5 minutes.
4. The egg protection system shall accommodate a minimum of two (2) eggs.

Rocketry:

Previous rocketry launches conducted by members of the group have included an H-class motor without a major payload involved. The primary target for previous launches was to successfully launch and return in a flightworthy condition, without major damage. The major challenge with this competition is the added factor of fragile payload that must be protected during launch. This added challenge poses added design and stability factors when it comes to the airframe of the rocket. Previous obstacles groups have faced when getting go/no go permission for launch include lack of safety capability due to an unknown descent velocity most likely due to the weight carried by a large payload. Also, having the ability to install and remove the payload within a time limit poses a challenge that can affect overall structure and capability. Most rockets carrying an H or higher class motor tend to provide enough thrust for a high apogee; however, adding additional weight limits those capabilities and requires restructuring of the payload to meet a required altitude.

Carrying the payload will be a Level 1 class rocket. The National Association of Rocketry and the Tripoli Rocketry Association set the standards for rocketry. There are 4 levels of rocket motors and rocketry certification. Level 0 (beginner) includes ½ A through G motors, Level 1 (intermediate) includes H and I motors, Level 2 (advanced) includes J, K, and L motors, and Level 3 (high-power expert) includes M, N, and O motors. The priority for level 1 rocketry certification is safe recovery of the rocket, ensuring it is able to re-launch. Below is the list of NAR Certification Checklist [3] which the project vehicle will be based on:

1. Has the rocket model that is being used for the certification attempt been built by the applicant requesting certification?
2. Is the nosecone or payload shoulder sufficiently tight to prevent drag separation? The nosecone or payload should not wobble side to side or separate from its own weight. Is a

vent hole needed to relieve pressure for high altitude flight? Do stage couplers fit snugly to prevent bending or separation during flight? Is the body tube thickness adequate to withstand high power flight (typically .050 inch walls or thicker)? Is there pre-existing damage which may weaken the model structure (e.g. tube crimps)? Are screws and fasteners tight, if used?

3. Are the rail buttons properly sized (for 1010 rail or larger), positioned and aligned correctly, and securely fastened to the airframe? For launch lugs, are they properly sized for the model (typically 1/4 inch dia. or larger), positioned and aligned correctly, and securely fastened to the airframe (taped on lugs are not permitted).

4. On cluster models, are the spaces between the motor tubes filled to prevent ejection pressure leakage? If mixing black powder and composite motors, does the modeler assure composite motor ignition before black powder motor ignition (composite motors ignite more slowly than black powder motors)? If the cluster model is not using all of its motors, are the unused motor tubes plugged to prevent ejection blow-by?

5. If a Level 1 Certification is being attempted, does the rocket model contain at least one H or I impulse motor?

6. Is (are) the motor(s) sufficient to safely fly the model? Use motor manufacturer's recommendations or recommended motor lists for similarly sized models as a starting point (Also consider, model weight, configuration, and finish when evaluating motor capabilities). Is (are) the motor(s) either NAR, Tripoli or CAR certified? Motors must be currently certified to be used.

7. Is (are) the rocket motor(s) firmly restrained in the model? Check for engine mount integrity to prevent a "fly through" (Is a thrust ring used?). Check for a motor hook or similar motor restraint. Carefully check taped or friction fit motors for tightness. Ask the modeler what adhesives were used during assembly. Are clusters wired in parallel?

8. If electronics are used, is the battery secured against "g" loads? Will electrical connections fail or loosen from acceleration forces? Will igniters stay fully inserted in rocket motors during boost? Is the user protected against inadvertent operation, e.g. is the circuit remotely armed, are safety switches present, is an armed status indicator used (visual or audible)? Does the modeler have a checklist or reminder to arm or operate the system prior to flight?

9. If radio control is used for flight functions (e.g. recovery), is the operating frequency in the 27, 50, 53, or 72 megahertz bands? Use of 75 megahertz for flight functions is not permitted. Is the antenna protected from breakage (not flopping freely)? Did the operator range check their equipment?

10. Are the fins fully secured to the model? Check for looseness or cracking at the fin to body tube junction. "Thru the wall" construction is recommended for high power models. Is the fin material compatible with the motor thrust range (1/8 inch minimum plywood is recommended for high power models)? Ask the modeler how their fins are mounted, what adhesives were used (epoxy is preferred), and what fin material was used. Are the

fins mounted parallel to the roll axis of the model? Are any warps present which may cause erratic flight?

11. Is the model stable? If stability is in doubt, require proof of the CG and CP locations (remember CG should be forward of the CP by approximately 1.0 body tube diameters). Ask the modeler to show the CG and CP locations and how they were determined. Verify that the modeler shows the CG with the motor(s) intended for flight and not a smaller motor or fewer motors (clusters). Ask the modeler to show CG and CP for the complete model and less each stage for a staged model. Require evidence of CP calculations if further doubt exists

12. Is the model in compliance with the FAA Certificate of Waiver or Authorization (COA)? Verify compliance by comparing model weight and power with charts/tables (if available) or by calculation. Ask the modeler what the expected performance is and how this determination was made (e.g. computer simulation, similar models).

13. Does the recovery system being used follow the requirements of an Active Recovery deployment system required for certifying? Inspect the recovery system. Verify that the shock cord is not cut or frayed and is free of burns. Are the shock cord mounts securely mounted to the model? Are sharp edges present which may cut shock cords, parachute risers, and suspension lines? Is hardware, e.g. swivels, screw eyes, sufficiently strong to withstand recovery loads. If required, perform a pull test on the recovery system. Is parachute protection (e.g. wadding) adequate? Check for parachute damage, e.g. tears, burns, which may spread during recovery.

The standards developed above by the National Association of Rocketry will be the benchmark for the project's launch vehicle safety standards and reusability standards.

List of engineering design requirements:

1. Launch vehicle apogee **shall** exceed 999 ft, while not exceeding 1500ft
2. Launch vehicle **shall** land within 800 ft of the launch pad
3. Descent rate **shall** not exceed 25 ft/s during the last 5 seconds of flight
4. Launch vehicle **shall** be recovered in proper flightworthy condition
5. Launch vehicle **shall** contain data logging equipment to record altitude and velocity
6. Payload **shall** carry at least 5 oz. of water and at least 2 eggs with room for additional eggs.
7. Payload system **shall** be designed in a way so loading/unloading of materials is done within a 5 minute time span
8. Overall launch vehicle propulsion system impulse **shall** be ≤ 320 Ns

2.3. Requirements Matrix

Cells will be numbered based on needs, wants, and requirements listed above in sections 2.1 and 2.2. Importance will be ranked as 1 being the most important and 3 being the least important.

		Engineering Design Requirements							
		1	2	3	4	5	6	7	8
User Needs	1	3	3	1	2	3	3	3	3
	2	3	1	3	3	3	3	3	3
	3	3	3	1	1	3	3	3	3
	4	2	2	1	1	3	3	3	2
	5	3	3	3	3	3	1	3	3
	6	3	3	3	3	3	2	1	3
	7	3	1	1	3	3	3	3	3
	8	3	3	3	3	1	3	3	3
	9	3	3	3	3	1	3	3	3
	10	1	3	3	3	3	3	3	1
User Wants	11	3	1	1	3	3	3	3	3
	12	3	3	2	3	3	1	3	3
	13	3	3	3	3	2	3	3	3
	14	3	2	2	3	3	3	3	3
	15	3	3	3	3	2	3	3	3
	16	3	3	3	3	2	3	3	3
	17	3	3	3	3	2	3	3	3

Figure 1: Requirement Matrix depicting importance of user needs and user wants in relation to Engineering design requirements

2.4. Critical Performance Parameters

The critical performance parameters that shall dictate a successful system design are laid out by the University of Central Florida, the primary stakeholder of the CRASH competition. These specifications set the standard for what shall or shall not be deemed a valid entry.

CRASH Challenge Rules

- Launch vehicle apogee shall exceed 999 feet AGL (above ground level) and shall not exceed 1500 feet AGL.
- Launch vehicles shall land no more than 800 feet from the launch pad.
- A vehicle decent rate exceeding 25 feet per second during the final 5 seconds of flight will not be scored.
- The launch vehicle shall be recovered in flightworthy condition. Post launch flight worthiness will be verified via a contest judge
- Launch vehicles shall carry data logging equipment capable of recording altitude over time and decent velocity over time.
- Launch vehicles shall carry a redundant data logging system capable of recording apogee. The redundant system shall be self-contained meaning it does not require an external battery.
- Launch vehicles shall carry no less than 5 ounces of water and should be capable of carrying additional water.
- Water carrying vessels should be easy to access, able to be loaded or unloaded from the vessel in 5 minutes or less, and capable of loading and unloading water for purposes of evaluation by competition judges.
- Water carrying vessels shall be inspected pre and post launch by a contest judge to validate and verify the quantity of water onboard the vessel and verify no leaks are present. Leaking vessels will not be awarded credit.
- Launch vehicles shall carry no less than 2 eggstronauts and should be capable of carrying additional eggstronauts. Eggstronauts should be easy to access, able to be loaded or unloaded from the launch vehicle in 5 minutes or less, for purposes of evaluation by competition judges. No credit will be given to eggstronauts that sustain visible damage during a post flight inspection by a contest judge.
- The sum of all launch vehicle propulsion systems shall contain no more than 320

Newton seconds of total impulse.

- Teams must follow all NAR (National Association of Rocketry) and SRA (Spaceport Rocketry Association) safety standards and guidelines.
- All “Sparky” motors are not permitted to be used.
- At least 1 team member must be NAR or TRA certified Level 1 or greater to purchase and operate Level 1 propulsion options. Propulsion systems may not utilize motors rated greater than Level 1.

2.4.1 Preliminary Values

With these exact specifications in mind, the goal of the competition is to optimize the size of the payload while still meeting all of the above criteria. To do so with minimal risk, it will be preferable to target a middle range of values for preliminary analysis. The main categories where this is applicable are the target apogee, landing distance, descent rate, payload capacity, and total impulse.

For the apogee, the preferred target value will be the median elevation permitted, or 1250 feet. This allows for the widest range of acceptable performance, ±250 feet.

For landing distance, there is a large variability in the factors which are capable of affecting where the vehicle may touch down, such as the weather, therefore targeting the closest reasonable distance at 600 feet is preferred to ensure the vehicle lands within bounds with room for error. From prior experience launching at the launch site, 600 feet is deemed to be a reasonable target to avoid natural obstacles such as tall bush and heavy tree line. This also takes into account inclement weather conditions including wind gusts that may interfere with the landing target.

Minimizing the descent velocity is critical for ensuring safety of the payload. Therefore it shall be best to target less than 25 ft/s. Given that the weight of the launch vehicle will be large, the descent velocity in this case is targeted to be 18 ft/s.

The main objective of the CRASH competition is to maximize payload capacity while being within the rules set by the stakeholder. Optimizing the balance between payload capacity and mass is critical to performing well, therefore having the maximum capacity, while remaining large enough to stay within acceptable performance parameters, will be the primary goal for winning the competition. On top of this criteria, it is also crucial that the payload component of a minimum five ounces of water and two eggs be loaded and unloaded within a five minute time frame, so creating ease of access to the payload is important.

The total impulse required will vary based upon the total mass of the vehicle, it is important to tune this value such that the vehicle will stay within the acceptable range of other performance values. With this in mind, the launch vehicle would benefit from the maximum allowable impulse of 320 Newton-seconds to achieve the targeted altitude.

Proper flightworthy condition post-flight is a requirement that is overseen by standards set by NAR's high powered rocketry certification criteria. Overall, it is critical that this launch vehicle must complete flight in a safe and controlled manner.

2.5. Engineering Requirements

Components and performance requirements specified below will be inspected, analyzed, or tested to ensure proper design, manufacturing, and launch of the vehicle. All requirements shall be found satisfactory in order to be approved for launch to ensure full safety and capability. Each will also be separated and denoted by an assigned code and section:

Engineering Design Requirements - EDR 1.0
Aerostructures Requirements - AERO 2.0
Propulsion Requirements - PROP 3.0
Recovery Requirements - RCOV 4.0
Payload Requirements - PALD 5.0
Avionics Requirements - AVIO 6.0

Table 1: Engineering Requirements

Ref #	Requirement	Target Values	Verification Method
1.0 Engineering Design Requirements			
EDR - 1.1	Launch vehicle apogee **shall** exceed 999 ft, while not exceeding 1500 ft	1100 ft	Analysis
EDR - 1.2	Launch vehicle **shall** land within 800 ft of the launch pad	600 ft	Analysis
EDR - 1.3	Descent rate **shall** not exceed 25 ft/s during the last 5 seconds of flight	18 ft/s	Analysis
EDR - 1.4	Launch vehicle **shall** be recovered in proper flightworthy condition	Passes NAR Certification Standards	Inspection
EDR - 1.5	Launch vehicle **shall** contain data logging equipment to record altitude and velocity	Passes	Test
EDR - 1.6	The Payload **shall** carry at least 5 oz. of water.	5 oz.	Inspection
EDR - 1.7	The Payload **shall** carry at least 2 eggs.	12 eggs	Inspection
EDR - 1.8	Payload system **shall** be designed in a way so loading/unloading of materials is done within a 5 minute time span	4 minutes	Test
EDR - 1.9	Overall launch vehicle propulsion system impulse **shall** be ≤ 320 Ns	236 N*s	Analysis
2.0 Aerostructures Requirements			
AERO - 2.1	The target nose cone **shall** maintain an elliptical shape.	Elliptical shape	Inspection
AERO - 2.2	Nose cone **shall** be fully intact pre-flight, free of any material loss, denting, and cracking.	Passes NAR Certification Standards	Inspection
AERO - 2.3	Connection to the body tube **shall** be fixed and rooted under pressure.	Passes NAR Certification Standards	Inspection
AERO - 2.4	Target body tube diameter **shall** accommodate the size of the payload protection system.	4 in	Inspection

Ref #	Requirement	Target Values	Verification Method
AERO - 2.5	Body tube thickness **shall** be ≥ 0.050 inch.	0.05 in	Inspection
AERO - 2.6	The body tube **shall** have at least 2 rail buttons securely fastened with a ¼ inch diameter.	Quantity: 2 Size: 0.25 in dia.	Inspection
AERO - 2.7	The body tube **shall** be fully intact, free of any cracks, bubbling, or crushing.	Passes NAR Certification Standards	Inspection
AERO - 2.8	The fin quantity **shall** be greater than 2.	4 Fins	Inspection
AERO - 2.9	Fins **shall** be fully intact pre & post flight with a strong connection to the body tube and experience no bending, warping, extensive crack propagation, or major chips that could affect the launch vehicle's stability. Small chips, scratches, or dents along fin's edge are permissible as long as the overall integrity is not at risk.	Passes NAR Certification Standards	Inspection
AERO - 2.10	Bulkheads & motor mounts **shall** be fixed and intact; free of any cracking, denting, or any other types of deformation.	Passes NAR Certification Standards	Inspection
AERO - 2.11	Center of Gravity (CG) **shall** be at least one body diameter in front of its Center of Pressure (CP).	Passes NAR Certification Standards	Inspection
AERO - 2.12	Launch vehicle stability **shall** fall between the range of 1.5 - 2.0. Anything outside of this range may lead to failure.	2	Inspection
AERO - 2.13	Entirety of model **shall** be compliant with the FAA Certificate of Waiver or Authorization (COA).	Passes FAA Regulations	Inspection
AERO - 2.14	Aerostructure components **shall** be made up of materials that withstand the force caused by the propulsion system.	Passes NAR Certification Standards	Inspection

Ref #	Requirement	Target Values	Verification Method
AERO - 2.15	Aerostructure components shall withstand ignition conditions.	Passes NAR Certification Standards	Inspection
3.0 Propulsions Requirements			
PROP - 3.1	The motor **shall** be fully intact and able to be removed and replaced.	Passes NAR Certification Standards	Inspection
PROP - 3.2	Airframe and motor mount **shall** be free of extensive scorching or burning.	Passes NAR Certification Standards	Inspection
PROP - 3.3	Motor **shall** be securely fastened with the motor mount pre & post flight.	Passes NAR Certification Standards	Inspection
PROP - 3.4	Motor **shall** not be in an impulse class higher than H.	H class	Inspection
4.0 Recovery Requirements			
RCOV - 4.1	Target descent velocity **shall** be less than 25 ft/s.	18 ft/s	Analysis
RCOV - 4.2	Target landing position of the rocket **shall** be 800 ft or less from the initial launch site.	600 ft	Analysis
RCOV - 4.3	Parachutes **shall** be fully intact and untangled, free of any holes, shredding, and any damage to the suspension lines.	Passes NAR Certification Standards	Inspection
RCOV - 4.4	Shock cord **shall** be fully intact; free of any fraying, extensive burns, and any loose knots/ties.	Passes NAR Certification Standards	Inspection
RCOV - 4.5	Deployment of recovery systems **shall** result in no damage to the airframe such as zippering on the body tube.	Passes NAR Certification Standards	Inspection
5.0 Payloads Requirements			
PALD - 5.1	Payload protection components (vibration and impact dampers) combined **shall** weigh no more than 7 oz.	7 oz	Inspection

Ref #	Requirement	Target Values	Verification Method
PALD - 5.2	Target volume for water payload **shall** be 5 fl oz.	5 oz	Inspection
PALD - 5.3	Target quantity for “eggstronaut” payload **should** be 12 “eggstronauts.”	12 eggs	Inspection
PALD - 5.4	The egg protection system **shall** reduce vibration intensity to less than or equal to 7.5 m/s^2.	7.5 m/s^2	Test
PALD - 5.5	The egg protection system **shall** limit impact forces to less than 35N.	25 N	Test
PALD - 5.6	The egg protection system **shall** allow for eggs to be loaded and unloaded within 5 minutes.	5 minutes	Test
PALD - 5.7	The egg payload section **shall** weigh no more than 30 oz.	30 oz	Inspection
PALD - 5.8	Egg payload location **shall** be consistent compared to pre/post flight.	Passes	Inspection
PALD - 5.9	Holding vessels for eggs **shall** be free of cracks, dents, leaks, holes, or bending. Minor cracks and dents may be permissible as long as the integrity of the structure is not compromised and the fragile cargo is unharmed.	Passes	Inspection
PALD - 5.10	The water payload section **shall** weigh no more than 7 oz.	7 oz	Inspection
PALD - 5.11	Water payload location **shall** be consistent compared to pre/post flight.	Passes	Inspection
PALD - 5.12	Holding vessels for water **shall** be free of cracks, dents, leaks, holes, or bending. Minor cracks and dents may be permissible as long as the integrity of the structure is not compromised and the fragile cargo is unharmed.	Passes	Inspection

Ref #	Requirement	Target Values	Verification Method
6.0 Avionics Requirements			
AVIO - 6.1	The avionics bay **shall** weigh no more than 20 oz when fully configured.	15 oz	Inspection
AVIO - 6.2	The avionics system **shall** record flight data at a minimum frequency of 10 Hz for post-flight analysis.	20 Samples per second	Test
AVIO - 6.3	The avionics system **shall** be capable of collecting data for a maximum of 5 minutes of flight.	20 minutes	Test
AVIO - 6.4	The avionics system **shall** operate on a nominal voltage of 9 Volts.	9 Volts	Test
AVIO - 6.5	The avionics system **shall** not exceed an amperage load of 500 mA at any point during operation.	300 mA	Test
AVIO - 6.6	The avionics system flight data **shall** be able to be extracted directly from the flight controller.	Passes	Test
AVIO - 6.7	The avionics system **shall** include a backup recording device for flight characteristics.	Passes	Inspection
AVIO - 6.8	The avionics bay **shall** integrate vent ports to accommodate onboard barometers.	Passes	Inspection
AVIO - 6.9	The avionics system **shall** employ a dual-deployment recovery setup.	Passes	Inspection
AVIO - 6.10	The avionics system **shall** initiate the deployment of a recovery device at a minimum of 500 feet of altitude of flight descent.	500 feet	Test
AVIO - 6.11	The avionics system **shall** allow for the use of e-match pyrotechnic devices for recovery equipment initiation.	Passes	Inspection
AVIO - 6.12	The avionics system **should** incorporate an onboard camera facing outward from the vehicle.	Passes	Inspection

Ref #	Requirement	Target Values	Verification Method
AVIO - 6.13	The avionics system **should** capture and store camera footage onboard the vehicle.	Passes	Test
AVIO - 6.14	The avionics system onboard footage **should** be able to be extracted via microSD.	Passes	Test
AVIO - 6.15	The avionics system **shall** be armed externally while the vehicle is on the launch pad.	Passes	Test
AVIO - 6.16	The avionics system **shall** employ the use of an arming switch accessible through vent ports located on the side of the avionics bay.	Passes	Test
AVIO - 6.17	Onboard electronics **shall** be properly connected, free of any burns or damages leading to the electronics being unreadable/unusable before or after flight.	Passes NAR Certification Standards	Inspection

3.0 References (AIAA Format)

[1] Food and Agriculture Organization of the United Nations, “Packaging and Transportation of Eggs,” FAO Agricultural Services Bulletin 150, 2003. [Online]. Available: https://www.fao.org/4/y4628e/y4628e05.htm

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[3] National Association of Rocketry, “Level 1 High Power Rocketry Certification Procedures,” revised January 1, 2022. [Online]. Available: https://www.nar.org/Level1HPRCertificationProcedures